D Stronger Failure Concentration with Larger K

To supplement Table 1, we compare TV distance for each parameter dimension with increasing K. The distance is measured between the normalized failure distribution per parameter dimension and the uniform baseline; we report the three parameter dimensions with the largest distances. We observe that increasing K leads to larger distances across models and templates. This implies that with larger K, the failure distribution has shorter tails (fewer inconsistent failures) and probability mass concentrates more on the dominant failure-inducing inputs. This empirically supports the efficiency gains that grow with K in Table 2.

Qwen2.5-Math-7B-Instruct

	$K = 4$			$K = 8$			$K = 16$		
Template	Top1	Top2	Top3	Top1	Top2	Top3	Top1	Top2	Top3
0	0.822	0.705	0.400	0.884	0.749	0.433	0.899	0.749	0.509
1	0.629	0.485	0.477	0.857	0.713	0.710	0.914	0.833	0.750
2	0.424	0.213	0.069	0.495	0.234	0.022	0.538	0.249	0.019
3	0.230	0.218	0.184	0.364	0.296	0.219	0.472	0.362	0.310
4	0.324	0.173	0.126	0.347	0.229	0.154	0.339	0.264	0.185
5	0.752	0.446	0.266	0.799	0.599	0.559	0.939	0.866	0.799
6	0.835	0.623	0.333	0.862	0.708	0.269	0.878	0.752	0.246
7	0.682	0.588	0.380	0.773	0.696	0.528	0.797	0.748	0.556
8	0.850	0.750	0.524	—	—	—	—	—	—

gpt-oss-20b-low

	$K = 8$			$K = 16$			$K = 24$		
Template	Top1	Top2	Top3	Top1	Top2	Top3	Top1	Top2	Top3
0	0.708	0.621	0.394	0.900	0.800	0.695	0.933	0.850	0.729
2	0.550	0.390	0.041	0.648	0.459	0.053	0.699	0.541	0.052
4	0.419	0.292	0.223	0.552	0.396	0.335	0.631	0.508	0.385
7	0.694	0.412	0.390	0.836	0.555	0.468	0.895	0.633	0.537
8	0.680	0.541	0.300	0.732	0.590	0.413	0.760	0.599	0.487

Gemini 2.5 Flash Lite

	$K = 8$			$K = 16$			$K = 24$		
Template	Top1	Top2	Top3	Top1	Top2	Top3	Top1	Top2	Top3
0	0.739	0.664	0.118	0.829	0.792	0.237	0.867	0.850	0.267
2	0.696	0.691	0.032	0.848	0.750	0.062	0.907	0.750	0.080
4	0.201	0.093	0.086	0.224	0.110	0.107	0.208	0.133	0.102
6	0.747	0.701	0.118	0.969	0.875	0.143	0.983	0.889	0.189

Table 3: Top-3 largest TV distances between the failure distribution and the uniform baseline, reported across models, templates, and majority-vote ensemble sizes K. — denotes model-templat pairs that have zero failures.

E Failure Histograms for All Parameters

Figure 6 shows parameter-wise failure histograms for all parameters, supplementing Figure 2 in Section 4. Parameters such as `Name`, `Drinks`, and `Locations` are roughly uniformly distributed among failures. In contrast, parameters such as `n` (first column), `frac2` (second column), and `x` (third column) show strong concentration of failures at a few specific values.

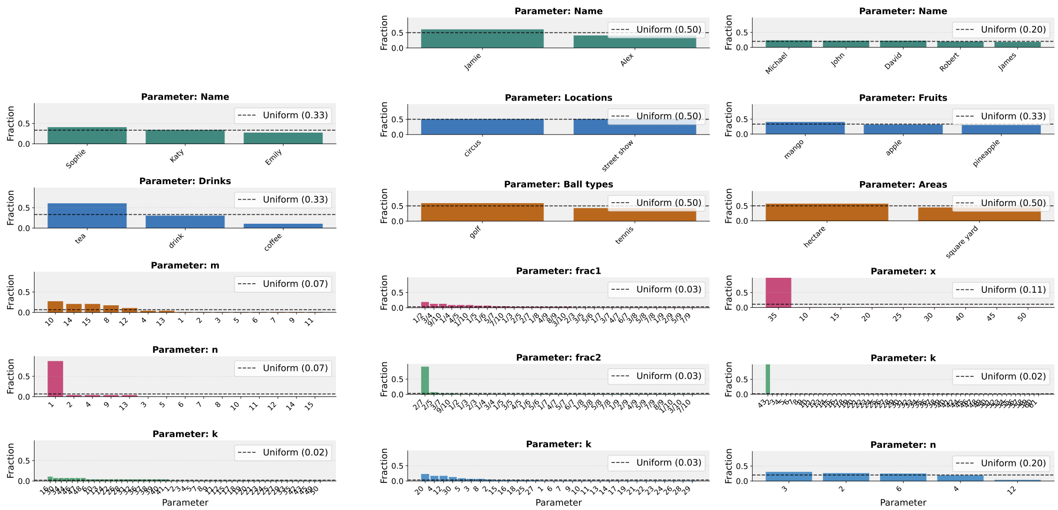

Figure 6: Parameter-wise failure histograms reveal strong concentration of errors.

F Table of Failure Probability Evaluation

This section provides the full table of estimated failure probabilities across models, templates, and K ($\lambda = 0.1$), corresponding to Figure 5.

ID	Estimate ($\hat{\mu}_Q$) $\pm$ Bounds	Estimate ($\hat{\mu}_Q$) $\pm$ Bounds	Estimate ($\hat{\mu}_Q$) $\pm$ Bounds
	Qwen2.5-Math-7B-Instruct		
	K=4	K=8	K=16
0	1.50e-03 $\pm$ 4.90e-05	8.98e-04 $\pm$ 3.03e-05	6.14e-04 $\pm$ 1.50e-05
1	1.79e-03 $\pm$ 9.20e-05	4.34e-04 $\pm$ 3.36e-05	2.16e-04 $\pm$ 1.79e-05
2	1.25e-01 $\pm$ 6.97e-04	9.11e-02 $\pm$ 5.73e-04	7.21e-02 $\pm$ 4.98e-04
3	4.36e-02 $\pm$ 4.87e-04	1.80e-02 $\pm$ 2.95e-04	9.29e-03 $\pm$ 1.94e-04
4	7.79e-02 $\pm$ 6.22e-04	5.62e-02 $\pm$ 5.21e-04	4.47e-02 $\pm$ 4.61e-04
5	2.95e-03 $\pm$ 9.20e-05	7.41e-04 $\pm$ 3.29e-05	2.03e-04 $\pm$ 1.62e-05
6	4.40e-03 $\pm$ 5.02e-05	3.50e-03 $\pm$ 4.16e-05	3.08e-03 $\pm$ 3.48e-05
7	2.28e-03 $\pm$ 6.37e-05	1.46e-03 $\pm$ 3.50e-05	1.14e-03 $\pm$ 2.13e-05
8	1.96e-04 $\pm$ 2.36e-05	0.0, 5.30e-06	0.0, 5.30e-06
	gpt-oss-20b-low		
ID	K=8	K=16	K=24
0	5.22e-04 $\pm$ 3.40e-05	2.54e-04 $\pm$ 1.90e-05	1.70e-04 $\pm$ 1.19e-05
2	4.87e-02 $\pm$ 3.55e-04	2.55e-02 $\pm$ 2.42e-04	1.89e-02 $\pm$ 1.98e-04
4	1.09e-02 $\pm$ 2.24e-04	3.71e-03 $\pm$ 1.15e-04	2.29e-03 $\pm$ 7.96e-05
7	1.42e-03 $\pm$ 6.18e-05	5.17e-04 $\pm$ 2.70e-05	3.42e-04 $\pm$ 1.92e-05
8	3.49e-03 $\pm$ 7.65e-05	1.61e-03 $\pm$ 4.93e-05	1.12e-03 $\pm$ 3.86e-05
	Gemini 2.5 Flash Lite		
ID	K=4	K=8	K=16
0	2.74e-03 $\pm$ 1.21e-04	7.18e-04 $\pm$ 3.63e-05	3.28e-04 $\pm$ 1.65e-05
2	9.53e-03 $\pm$ 1.66e-04	4.73e-03 $\pm$ 7.29e-05	3.35e-03 $\pm$ 4.06e-05
4	5.54e-02 $\pm$ 5.69e-04	3.37e-02 $\pm$ 4.42e-04	2.59e-02 $\pm$ 3.87e-04
6	8.70e-03 $\pm$ 2.22e-04	1.27e-03 $\pm$ 6.26e-05	9.53e-04 $\pm$ 1.37e-05

Table 4: Estimated failure probability across templates, models, and K at $\lambda = 0.1$. For all missing model-template pairs are those that have zero failures within our evaluation budgets, and they can be estimated via exact binomial bounds of (0, 5.30e-06).